



SPAWNER & POOLER

DEVELOPED BY



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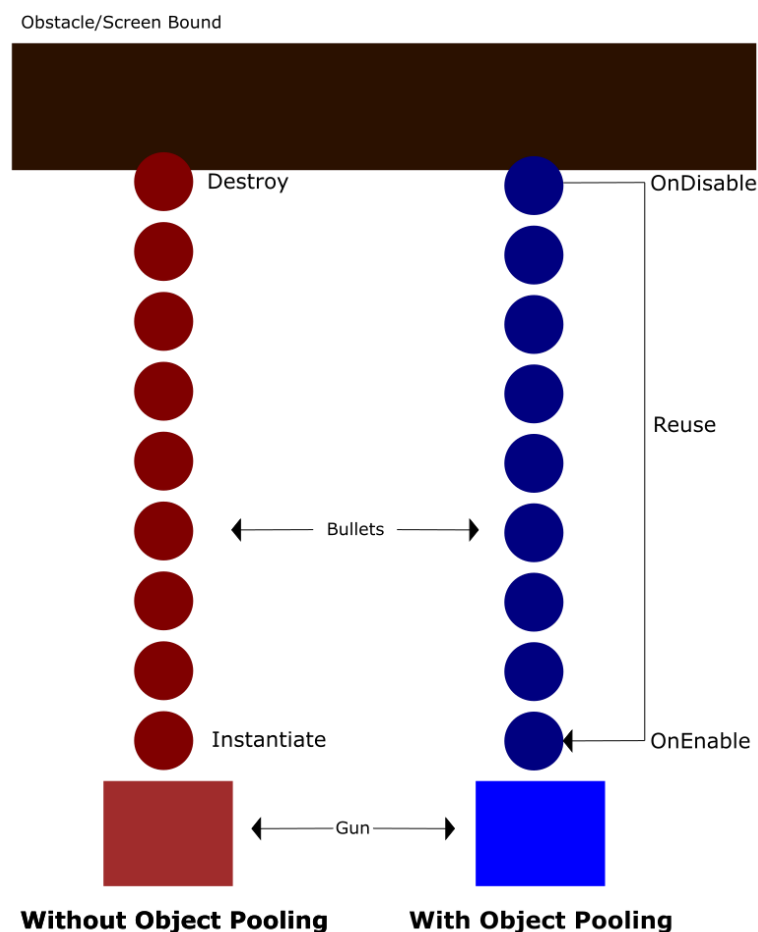
Concept of Object Pooling

When a gameobject is instantiated or destroyed at runtime there is a huge amount of CPU overhead which most programmers neglect. Allocation and deallocation of memory also get untidy to handle by Unity's garbage collector. Hierarchy window sometimes get unmanageable when there is a large bunch of instantiation and destruction of gameobjects.

To counter all these issues, Object Pooling can be done. Object pooling is to pre-instantiate required gameobjects and storing it in a pool. When needed a gameobject is returned from the pool. When needed to destroy it, it is deactivated and it still exists in the pool and can be reused when required again.

Endless path generation, firing bullets are the most common examples of object pooling. But it can be used for any gameobject that needs to be instantiated in the scene.

In the example below, it is shown how object pooling works.



Object Pooler

Access

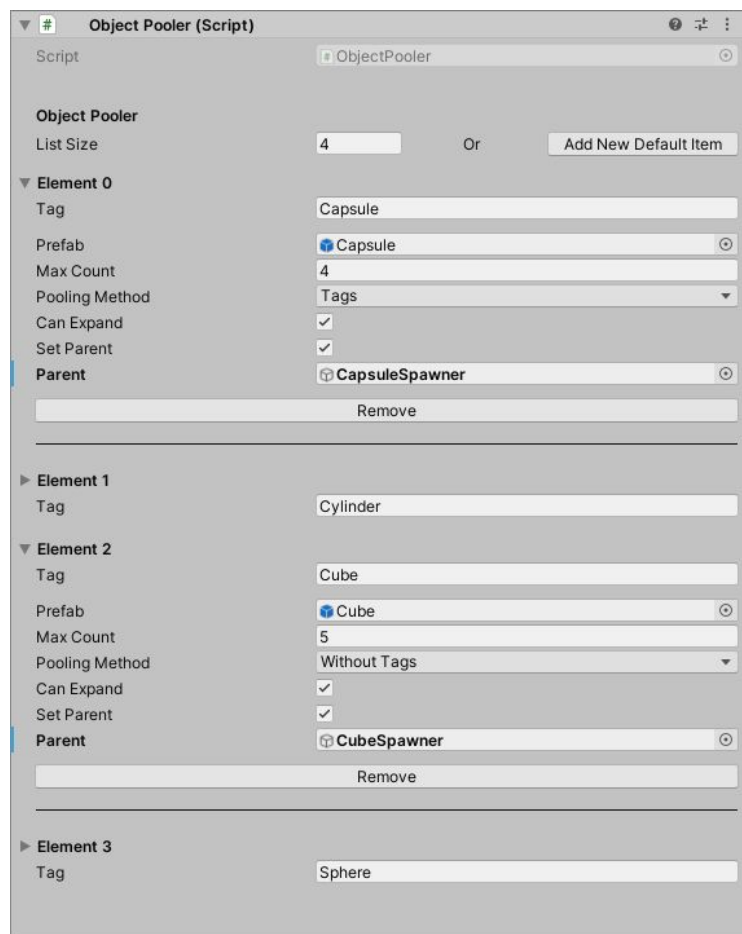
In the Project Folder, Goto Assets -> Spawner and Pooler -> Prefabs -> Drag ObjectPooler into Hierarchy

OR

Select any GameObject from the scene, Add Component -> Object Pooler

Tool

- **Tag:** every tag should be unique.
- **Prefab:** select a GameObject to pool.
- **Max Count:** number of GameObjects in the pool
- **Pooling Method:** Select the pooling method
 - **Tags:** Only the GameObjects active in the scene with the same Tag will be added to the pool
 - **Without Tags:** It Generates the pool at start
- **Can Expand:** whether the pool can be expanded, if no de-active gameobject is present
- **Set Parent:** whether the pooled GameObject should be child of the parent
 - **Parent:** select the Parent GameObject from the hierarchy to spawn the instantiated GameObject under the parent in the Hierarchy.
- **Remove:** to remove the pool element from the Object Pooler.



Scripting API

Spawning a Bullet

Function:

SpawnFromPool(string, Vector3, Quaternion) - Returns a GameObject from the pool.

Hint: Position and Rotation can also be manipulated in OnEnable() of the spawned prefab GameObject (Implementation is present in Sample Scene for prefabs Cubes and Cylinder)

```
using UnityEngine;
Using Indiemount.Pooler; //This namespace should be used to implement
Object Pooler Class

//This script is attached to an Empty gameobject which is also the spawn
point for the bullets
public class BulletSpawner : MonoBehaviour
{
    Transform spawnPoint;

    private void Start()
    {
        spawnPoint = gameObject.transform.GetChild(1).transform;
    }

    void Update()
    {
        if (Input.GetMouseButtonDown(0)) // When Left Mouse Button is
Pressed
        {
            //Accessing a GameObject From the Pool "Sphere"
            ObjectPooler.Instance.SpawnFromPool("Sphere",
spawnPoint.position, Quaternion.identity);
        }
    }
}
```

Sphere

```
using UnityEngine;

[RequireComponent(typeof(Rigidbody))]
public class Sphere : MonoBehaviour
{
}
```

```

float timer = 4;
bool shoot = false;

/// <summary>
/// Write the Function to Perform When this GameObject is Set to
Active from the Pool
/// </summary>
private void OnEnable()
{
    OnObjectSpawn();
}

/// <summary>
/// Write the Function to Perform When this GameObject is
Deactivated from the Pool
/// Mostly used to reset values
/// </summary>
private void OnDisable()
{
    //Your Code
}

public void OnObjectSpawn()
{
    timer = 4;
    shoot = true;
    Rigidbody rb = GetComponent<Rigidbody>();
    var dir = transform.position - Camera.main.transform.position;
    rb.AddForce(Camera.main.transform.forward * 2000,
ForceMode.Force);
}

private void Update()
{
    timer -= Time.deltaTime;
    if (timer <= 0)
    {
        //GameObject is Deactivated but still remains in the pool
        gameObject.SetActive(false);
    }
}
}

```

Wave Spawner

Access

In the Project Folder, Goto Assets -> Spawner and Pooler -> Prefabs -> Drag WaveSpawner into Hierarchy

OR

Select any GameObject from the scene, Add Component -> WaveSpawner

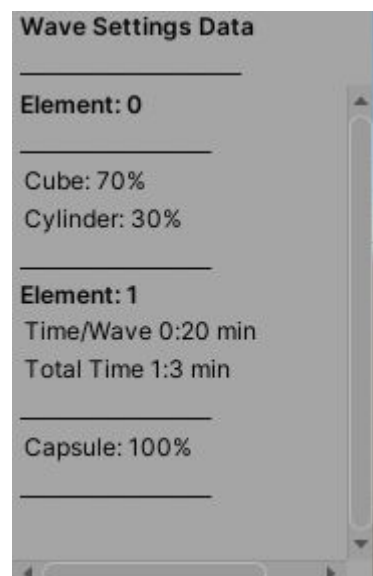
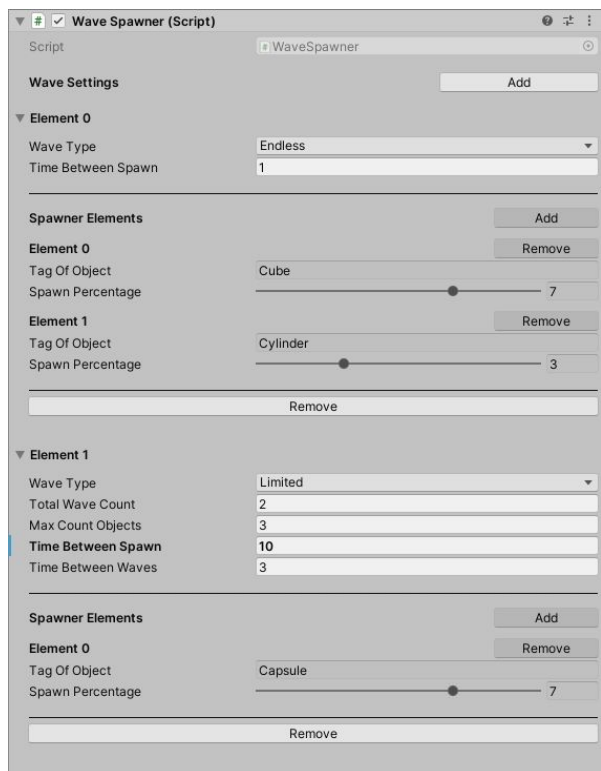
Tool

Wave Settings

- **Wave Type (Endless, Limited):**
Type of wave
 - **Endless:** Never ending waves. Mostly used for endless modes, where spawning stops only when player is dead
 - **Limited:** Only the specified amount of waves will be spawned
- **Total Wave Count:** total number of waves to be spawned
- **Max Count Objects:** number of GameObjects per wave
- **Time Between Spawn:** (Also Available in Endless Wave Type)
time difference between two spawn
- **Time Between Waves:** time difference between two waves

Spawner Elements

- **Tag Of Object:** Input Tag should be present in the Object Pooler
- **Spawn Percentage:** set percentage in-between 0-10. The calculated percentage will show in the Scene View at the Top Left Corner as Wave Settings Data. (Image on right, shows the Calculated data for the above Wave Settings)



Scripting API

Starting and Stopping Waves

Functions:

StartSpawning() - All the elements present in the Wave Settings gets started

StartSpawning(int) - Element at specified index in the Wave Settings gets started

StopSpawning() - All the elements present in the Wave Settings gets stopped

StartSpawning(int) - Element at specified index in the Wave Settings gets stopped

```
using UnityEngine;
using Indiemount.Spawner; //This namespace should be used to implement
Object Pooler Class

public class GameManager : MonoBehaviour
{
    void Start()
    {
        //Starting a Wave using the element index
        WaveSpawner.Instance.StartSpawning(0);
        WaveSpawner.Instance.StartSpawning(1);
    }

    void Update()
    {
        if (Input.GetKeyDown(KeyCode.S))
        {
            //Stopping all Waves
            WaveSpawner.Instance.StopSpawning();
        }
    }
}
```

Shape Spawner

Access

Goto Windows -> Shape Spawner

Shape Spawner inspector pops up. Place it where you want according to your convenience

Tool

Shape Spawn Settings

- **Spawn Prefab:** Select the prefab you want to spawn
- **Start Position:** Input the initial position of the first Spawner prefab
 - X 5 Y 10 Z 5
- **Padding:** distance between 2 spawned objects in X Y And Z axis separately.
 - X 20 Y 10 Z 10
- **Set Parent:** whether the pooled Gameobject should be child of the parent
 - ☒ **Parent:** select the Parent Gameobject from the hierarchy to spawn the instantiated Gameobject under the parent in the Hierarchy
 - Parent:** CubeSpawner
 - Layout:** Rectangle
 - Axis:** XYZ
 - Rows:** 5
 - Columns:** 4
 - Height:** 6
- **Pooler:**
 - ☒ **Add To Pooler**
 - Tag:** Box
 - ☒ **Can Expand**
- **Layout:** There are 3 available layouts (*Rectangle Triangle, Right Angle Triangle*). Select a layout to spawn in the desired shape.
- **Axis (X, XY, XYZ):** select the axis to spawn in the desired axis

The screenshot shows the 'Shape Spawner' inspector window. It has a title bar 'Shape Spawner' and a sub-header 'Shape Spawn'. The settings are organized into sections: 'Spawn Prefab' (set to 'Cube'), 'Start Position' (X: 5, Y: 10, Z: 5), 'Padding' (X: 20, Y: 10, Z: 10), 'Set Parent' (checked), 'Parent' (set to 'CubeSpawner'), 'Layout' (set to 'Rectangle'), 'Axis' (set to 'XYZ'), 'Rows' (5), 'Columns' (4), 'Height' (6), 'Pooler' (checked), 'Add To Pooler' (checked), 'Tag' (set to 'Box'), 'Can Expand' (checked), a 'Generate' button, 'Save System' (File Name: 'Box', Save button), and 'Load Asset' (set to 'Box (ShapeSpaw)', Load button).

Pooler

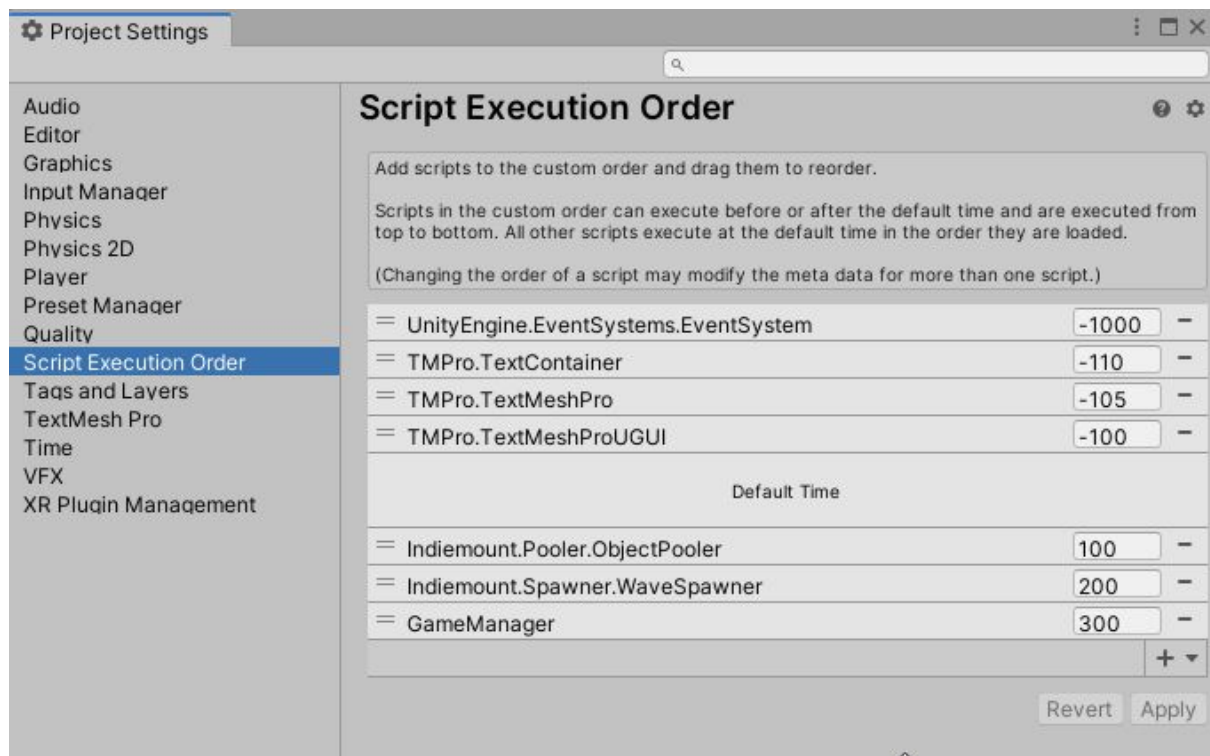
- **Add to Pooler:** if you set to True, the instantiated Gameobject will be added to the Object Pooler.
- **Tag:** every tag should be unique.
- **Can Expand:** whether the pool can be expanded, if no de-active gameobject is present
- **Generate:** To generate the Gameobject Accord to the desired settings. I the Add to Pooler boolean is checked, the pooler settings is added to Object Pooler (if GameObjects are not added to Object Pooler and shows Null Reference Error, Play the scene once and return, and Press Generate Button again)

Save System

- **File Name:** Input the file name you want to save as. If already present then it will replace the settings. And if not present New file will be created.
 - **Load Asset:** select the ShapeSpawnerSave file which you want to load and press "Load".
- ☐ Data Stored from ShapeSpawner is saved at location
Assets/Resources/Data/ShapeSpawner as ShapeSpawnerSave.asset

Script Execution Order

Firstly, Pool of gameobjects are created. Then waves are generated. And at last, these waves and pools can be accessed. The command for execution of these Waves and accessing the Pooler element can be done from a custom script (in this case GameManager).



Contact Us

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